

Reference

**Modern electrical
machines**

Package information

Reference

Understanding the system

Modern electrical machines



The system consists of a number of 24V electrical machines, a control unit, software applications for driving the control unit and a set of worksheets. The photograph above shows these parts. They are:

- 1) The dynamometer and cradle which connects to the control unit using a 25way D-type lead
- 2) Balance
- 3) Motor under test - in this case a shunt wound motor
- 4) The control unit which connects to a PC using a USB lead.
- 5) A series wound motor
- 6) Single-phase AC induction motor
- 7) Brushless DC motor

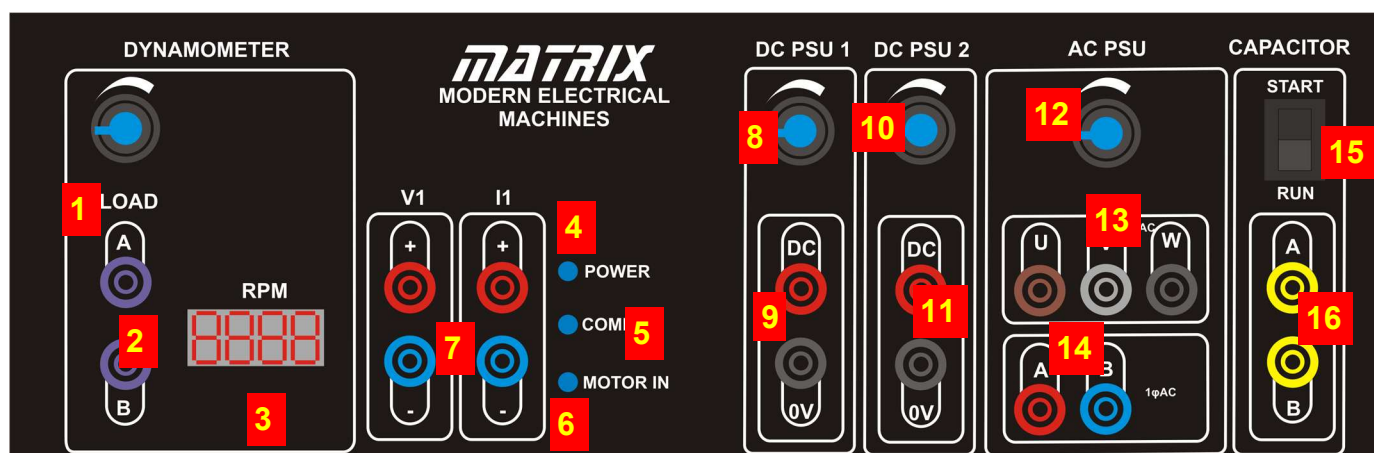
- 8) DC motor
- 9) Three-phase AC induction motor
- 10) The software application running on a PC

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Understanding the control unit

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Note - manual controls are overridden as soon as the unit is plugged into a computer.



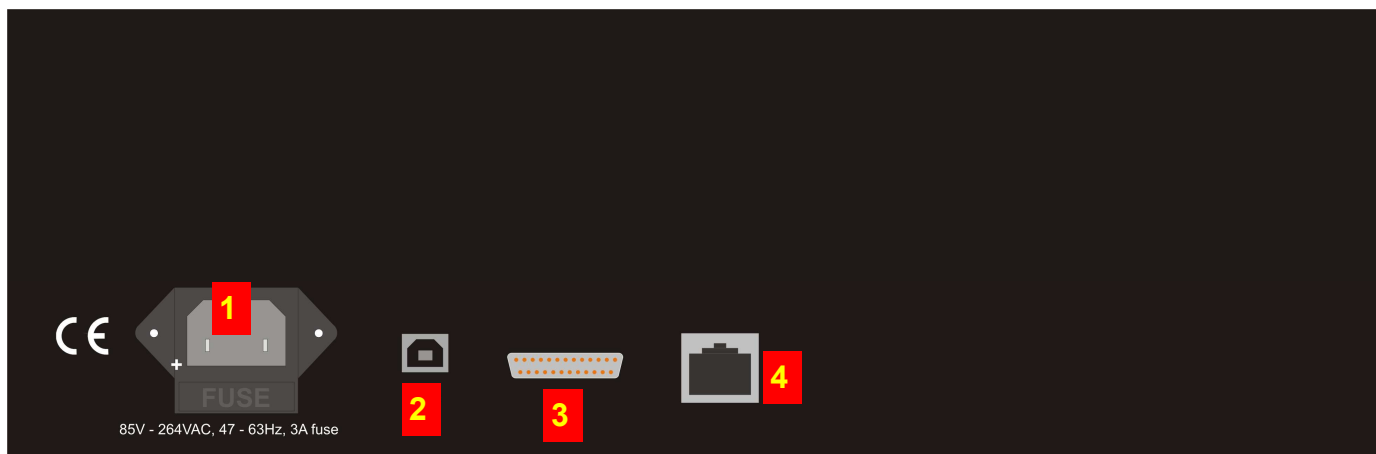
- 1) The dynamometer resistance: use this to control the effective resistance placed across the dynamometer: low electrical resistance means a large mechanical resistance, and high electrical resistance means low mechanical resistance
- 2) The dynamometer connections
- 3) The speed of the motor in Revolutions Per Minute - RPM.
- 4) The power LED which shows the control unit is powered up.
- 5) The COMMS LED which is lit when the PC software has communication with the control unit.
- 6) The 'Motor In' LED which indicates that a machine is physically connected to the dynamometer.
- 7) The internal ammeter and voltmeter connections
- 8) The DC 1 Supply output control.
- 9) The DC 1 connections.

- 10) The DC 2 Supply output control.
- 11) The DC2 connections.
- 12) The AC frequency control
- 13) The three-phase supply connections
- 14) The single phase supply connections
- 15) The capacitor mode selector switch: this controls the internal value of capacitor connected to the A and B terminals. There are two values: START, RUN.
- 16) Variable capacitance terminals.

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Understanding the control unit

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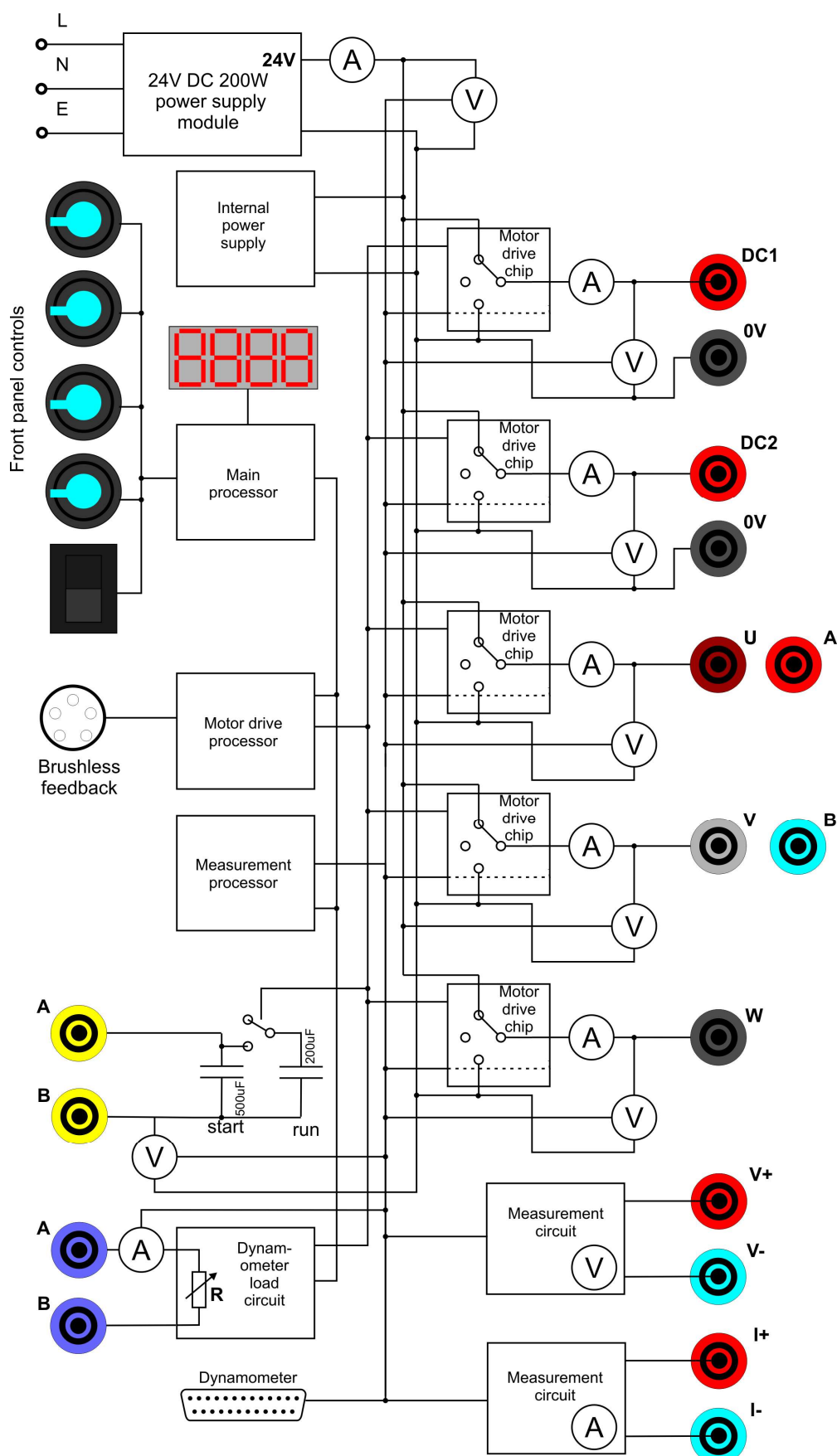


- 1) The Mains input plug - the unit takes 240V or 110V
- 2) The USB connector
- 3) A 25 way D-type connector which is used to connect the Control unit to the Dynamometer
- 4) RJ45 LAN/internet connection socket.

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Control unit schematic

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Control unit description

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Please refer to the block diagram on the previous page.

The control unit shipped with the Matrix Modern Electrical Machines system is one of the most up to date in the World. Inside the unit there are three separate processors: a main processor handling all user interface and communication tasks, a motor drive processor handling all the high current outputs and dealing with all waveform and timing systems, and a measurement processor. These three processors use buses to communicate between themselves and between the different electronic circuits in the unit.

A key feature of the control unit is that almost every quantity in the unit can be measured. On the schematic you can see that there are 16 separate ammeters and voltmeters in the system. This gives the instrumentation software and the user lots of options when displaying what is happening in an electrical machine system. The measurement of each quantity takes place thousands of times per second and can be processed by PC software applications to display a quantity or a waveform.

The motor drive chips are all low voltage drop FET based units. As you can see from the schematic all the outputs are digital 24V outputs. Simple pulse width modulation algorithms are used to vary the effective output power on the DC outputs. On the AC outputs Pulse width modulation is again used but with a more advanced pseudo-sine with third harmonic wave algorithm which varies the output power sinusoidally over the period of the output waveform. This technique is used by the more advanced motor controllers in industry. Older, and perhaps cheaper, motor driv-

er systems sometimes use a simple digital output waveform for driving three phase induction motors where the three outputs are simply digital outputs phase shifted by 120 degrees. This digital output is also available in some software applications shipped with the system so that students can investigate the efficiency of each method.

The motor driver chip outputs can be placed in one of three states: 24V, 0V and open circuit. This allows for different drive strategies when generating waveforms for driving motors. As part of the learning package students investigate PWM for driving DC motors, and Pseudo-sine waves for driving single phase AC motors. The unit also performs more complex six step pseudo-sine wave generation for driving the three phase motor and this can be examined with the internal multi channel oscilloscope as well as an external oscilloscope. To facilitate this when in AC mode a small pulse is given on the DC1 output which allows an external oscilloscope to be triggered to capture the various waveforms.

Note that most of the internal ammeters and voltmeters are on the Driver board in the unit. The topmost ammeter and voltmeter in the block diagram monitor power to the Driver board and this is displayed in the applications.

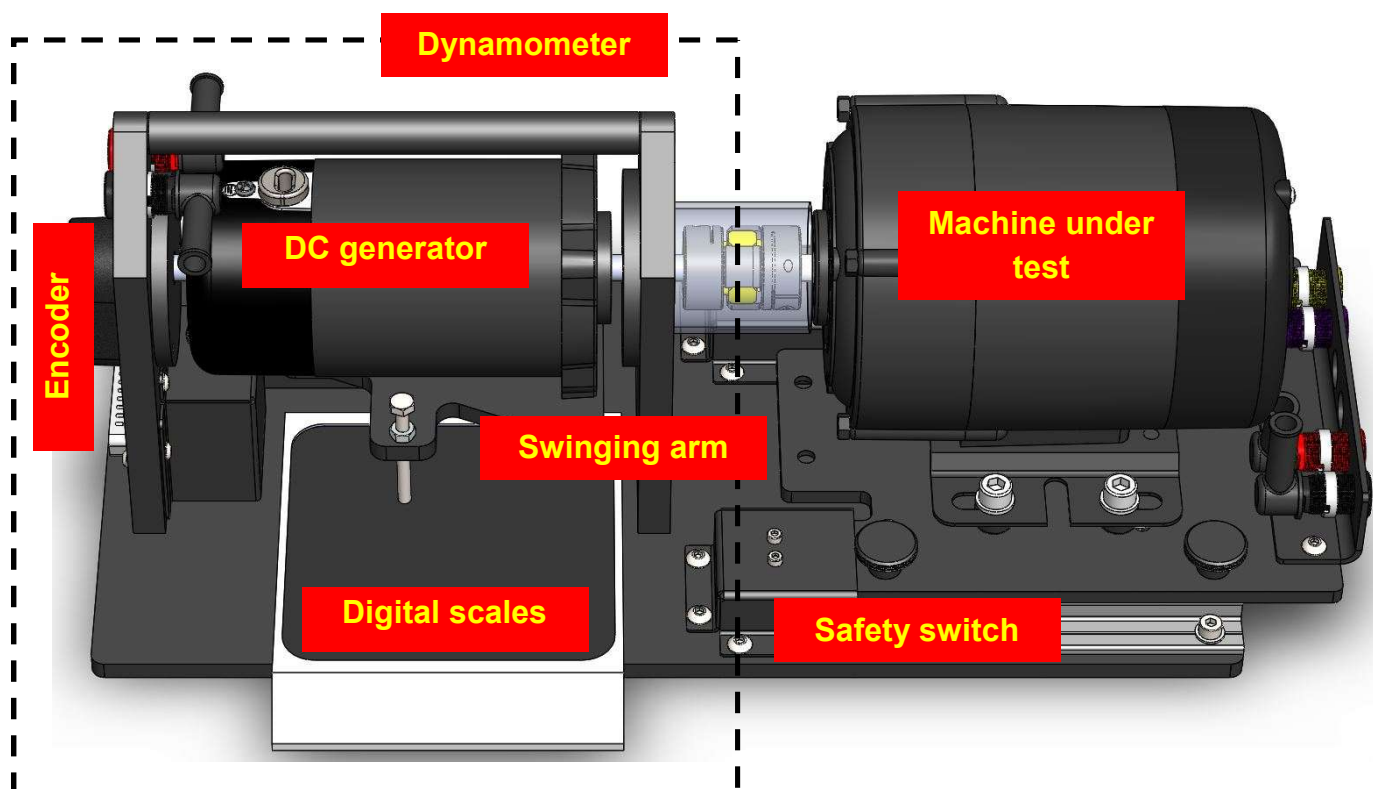
The unit includes two values of capacitor with a simple relay for start / run investigation for single phase induction motors.

The unit also includes a FET based load for the dynamometer whose effective resistance is controlled by software.

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Understanding the dynamometer

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A dynamometer is an instrument that directly measures the force produced and speed of a rotating machine (the motive force). Torque and power can be calculated from the measured force and speed. In our case the mechanical rotation is produced by one of the DC or AC electrical motors. This Dynamometer is an absorption type swing arm / balanced beam dynamometer, where the motive force is used to drive a DC generator with a variable resistance load attached. The resulting force can be measured in one of two ways:

- A digital scale (shown in the image above) can be used to measure the effective vertical force on the scale and it gives a reading in Kilograms. This can be converted into newtons and newton metres - see below.
- A load cell (behind the dynamometer and

not shown in the image above) connected to the control box gives the same information - give or take a few percent.

Note that the rotation of the dynamometer dictates whether the balance or the load cell is used. Controlling the direction of motion of each machine is discussed below.

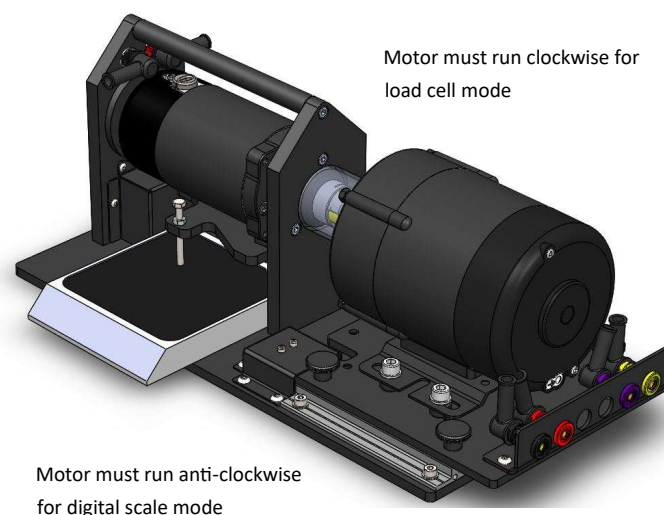
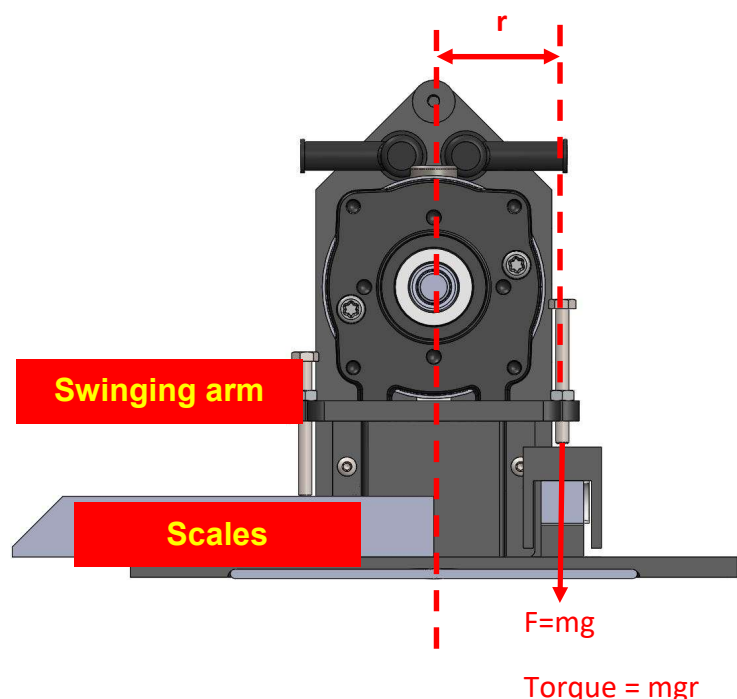
The encoder on the shaft of the dynamometer allows the control unit to detect the rotational speed. This is displayed on the control unit in revolutions per minute (rpm) and on the software application.

Machines under test are coupled to the dynamometer using a coupling which is housed in a plastic tube to prevent clothes and hair getting caught in the mechanism. Two microswitches are provided on the dynamometer - unless a machine under test is in place the control unit will not activate.

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Measuring torque

Modern electrical machines



The dynamometer rotates in the same direction as the machine under test. It provides a mechanical resistance for the machine under test. It also generates a voltage and - if a resistance is placed across the Dynamometer terminals - it produces a current.

If voltage and current flow then power is being generated - work is being done. The more power generated by the Dynamometer, the more mechanical power has to be generated by the machine under test.

We test electrical machines to understand their properties so we know when to use the various types of machine. The way we test them is to vary their speed and the mechanical power they need to generate. The dynamometer and the Control unit allow us to do this.

The Dynamometer is made up of a DC motor and a cradle. The DC motor also acts as a DC

generator. The cradle suspends the DC machine in two bearings that allow the body of the machine to rotate. The force on the load cell (or digital scale when the motor is running anticlockwise) creates a weight on the load cell or scale (which represents mass). If the swing arm is perfectly balanced and horizontal when it is at rest, the force produced on the load cell or digital scale is $\text{Force} = \text{mass} \times 9.81 \text{m/s}^2$. The torque can be calculated from the formula $\text{Torque} = \text{Force} \times \text{Distance}$. In this case, the distance is from the centre of rotation of the generator to the centre line of action of the swing arm. **For the Matrix dynamometer, $r = 38.12 \text{mm}$**

More discussion on calculations is given below.